

# Hankel-Structured Channel Estimation for Rydberg Atomic Receivers: Bounds, Effective-Rank Scaling, and Cross-Generator Validation

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**Abstract**—Rydberg atomic receivers measure field magnitude only, making multi-user channel estimation a biased phase-retrieval problem in which unstructured solvers are flat in array size: under the unconstrained row-wise parameterisation the Fisher information is block diagonal across receive elements, so normalised per-element information does not grow with  $N$  and exploiting aperture requires cross-element structure. We enforce the low-rank Hankel structure of the uniform-linear-array response inside an exact expectation–maximization Gerchberg–Saxton iteration, select the order from a held-out pilot residual, and benchmark against a Rician Cramér–Rao bound in which each magnitude measurement contributes rank-one information, together with a geometry-constrained bound on the  $3 \sum_k L_k$ -parameter manifold. The gain grows from  $-0.19$  to  $+2.85$  dB as  $N$  goes  $8 \rightarrow 32$  while the unstructured baseline moves by  $0.012$  dB. We then show the governing variable is not path count but effective rank relative to capacity, and use that relation to predict the gain on two channel-generation distributions it was not fitted to. Simulation only; the bounds do not govern biased estimators.

**Index Terms**—Rydberg atomic receiver, channel estimation, phase retrieval, Hankel matrix, constrained Cramér–Rao bound, effective rank.

## I. INTRODUCTION

ATOMIC receivers based on Rydberg states read out radio fields through an optical transition whose response is proportional to field *magnitude*, discarding phase [1], [2]. Multi-user channel estimation is therefore biased phase retrieval, and Gerchberg–Saxton-type alternating projection [3] is the natural solver; recent work unrolls it into a trained network [4], and a rank projection has been placed inside projected gradient descent for the same problem [5].

Unstructured solvers share a limitation that is easy to miss and, once seen, explains a great deal: **their accuracy does not improve as the array grows**. Under the unconstrained row-wise parameterisation of (1), row  $n$  of  $\mathbf{GS}$  enters only measurement row  $n$ , so the Fisher information is block diagonal across receive elements and the per-element estimation problem is unchanged by adding elements. Exploiting aperture therefore requires *cross-element* channel structure; HS-GS uses the low-rank Hankel structure induced by the geometric ULA model, which is one such structure rather than the only one.

On a uniform linear array (ULA) each propagation path contributes one spatial complex exponential, so the Hankel

lifting of every channel column is rank deficient [6], [7]. Enforcing that inside the iteration turns aperture into accuracy. But knowing *when* it applies has, until now, meant a qualitative appeal to sparsity, and that is unusable: a 16-path channel on a 32-element array is algebraically full rank yet has effective rank 8.73, and its lifting is compressible with nothing sparse about it.

## Contributions.

- 1) A benchmark, not just a baseline: a Rician Cramér–Rao bound in which each magnitude measurement contributes *rank-one* information, and a geometry-constrained bound in tangent-space form because the parameterisation is rank deficient. EM-GS numerically tracks the unconstrained bound to within 0.051 dB for  $\text{SNR} \geq 5$  dB, so the benchmark is informative.
- 2) The aperture-scaling result and its Fisher-information explanation.
- 3) Replacement of the path-count heuristic by  $r_{\text{eff}}/r_{\text{max}}$ , with the capacity  $r_{\text{max}} = \lceil N/2 \rceil$  derived.
- 4) Two cross-generator predictions, each registered before measurement. The relation is fitted on one channel-generation distribution and tested, without refitting, on two previously unseen ones — the clustered-propagation falsification test this work’s earlier stage identified as decisive.

## II. SYSTEM MODEL

An  $N$ -element ULA serves  $K$  single-antenna users over  $P$  pilots. With  $\mathbf{G} \in \mathbb{C}^{N \times K}$ , pilots  $\mathbf{S}$ , known reference beam  $\mathbf{B}$  and noise  $\mathbf{W}$ , the receiver observes

$$\mathbf{Z} = |\mathbf{GS} + \mathbf{B} + \mathbf{W}|, \quad \mathbf{W}_{np} \sim \mathcal{CN}(0, \sigma^2), \quad (1)$$

elementwise, and the modulus is never linearised. Here “biased phase retrieval” refers to the known reference-field offset  $\mathbf{B}$  and is distinct from the statistical estimator bias discussed in §IV. User  $k$ ’s channel is

$$\mathbf{g}_k = \sum_{\ell=1}^{L_k} \alpha_{k\ell} \mathbf{a}(\theta_{k\ell}), \quad [\mathbf{a}(\theta)]_n = e^{-j\pi n \sin \theta}, \quad (2)$$

so each column is a sum of complex exponentials in the element index.

**RSR convention, stated once.** The reference-to-signal ratio is defined against a *single* user. Since  $K$  users transmit at once, a nominal single-user RSR of 12 dB is  $12 - 10 \log_{10} K =$

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7.23 dB against the  $K$ -user total at  $K = 3$ ; 10 dB single-user is 5.23 dB. The empirically measured conversion is slightly larger than the nominal 4.77 dB — 4.85 dB over 300 realisations — which places the 12 dB case at 7.15 dB when measured rather than computed; the difference is finite-sample and we quote the nominal value throughout. All RSR values quoted here are single-user.

**Two averaging domains, kept distinct.** Within an operating point NMSE is pooled as a ratio of sums,

$$\text{NMSE} = 10 \log_{10} \frac{\sum_t \|\hat{\mathbf{G}}_t - \mathbf{G}_t\|_F^2}{\sum_t \|\mathbf{G}_t\|_F^2}, \quad (3)$$

never a mean of per-trial decibels: (3) estimates the ensemble quantity a Cramér–Rao bound bounds, whereas a mean-of-decibels is smaller by Jensen. Across SNR, the mean is of per-SNR decibel values, after conversion; linear NMSE spans three orders of magnitude, so a linear average across SNR is dominated by the lowest-SNR point and nearly blind to the high-SNR half. **Rule used throughout:** ratio-of-sums wherever a curve is compared to a bound; paired per-trial median with a bootstrap interval for paired contrasts. Both are reported; neither is used for the other’s job.

### III. HANKEL STRUCTURE AND THE HS-GS ESTIMATOR

For  $\mathbf{g} \in \mathbb{C}^N$  and pencil  $p$ , let  $\mathcal{H}_p(\mathbf{g})$  be the  $(N-p) \times (p+1)$  Hankel matrix. Under (2) its rank is  $\min(L_k, N-p, p+1)$ , so the largest expressible rank is

$$r_{\max}(N) = \max_p \min(N-p, p+1) = \lceil N/2 \rceil, \quad (4)$$

attained at  $p = \lfloor N/2 \rfloor$ . The ceiling, not the floor: at  $N = 7$  the maximum is 4, and the two expressions coincide only for even  $N$ . A constraint at  $r = r_{\max}$  is satisfied by every vector and carries no information, which fixes the scale any rank must be read against.

$K$  **cancels.** Unstructured,  $\mathbf{G}$  has  $2NK$  real degrees of freedom; under (2) it has  $3 \sum_k L_k$ . With  $L_k \equiv \bar{L}$ ,

$$\frac{3 \sum_k L_k}{2NK} = \frac{3K\bar{L}}{2NK} = \frac{3\bar{L}}{2N}, \quad (5)$$

independent of  $K$ . Since the projection acts per user column, this predicts  $K$ -invariance of the structural gain at fixed pilot adequacy  $P/2K$ .

**The estimator.** HS-GS interleaves, after every exact EM-GS update, a Cadzow projection  $\mathcal{P}_r(\mathbf{g}) = \mathcal{H}_p^\dagger(\Pi_r(\mathcal{H}_p(\mathbf{g})))$ , where  $\Pi_r$  truncates to  $r$  leading singular values and  $\mathcal{H}_p^\dagger$  averages antidiagonals. One application is *not* idempotent: it is one step of an alternating projection between the rank set and the Hankel set, not a projection onto their intersection, so the number of sweeps is part of the operator’s definition and is stated with every result below. The exact step is the EM-GS update throughout, so HS-GS and HS-EM-GS name the same estimator; we use HS-GS to match the released code.

**Order selection uses no oracle.** At each operating point a subset of pilots is held out, the estimator is run at reduced iteration count for every candidate  $r \in \{1, \dots, r_{\max}\}$ , and the  $r$  minimising the held-out residual is selected. The rule also *abstains*: when it selects  $r \geq r_{\max}$  the projection is the identity and HS-GS reduces exactly to EM-GS.

**The governing variable.** We index results by the Roy–Vetterli effective rank [8] of the noiseless column’s lifting,  $r_{\text{eff}} = \exp(-\sum_i p_i \ln p_i)$  with  $p_i = \sigma_i / \sum_j \sigma_j$ , taken relative to  $r_{\max}$ . Unlike  $L$ , this is a property of any channel rather than of one generator, and it is what makes the cross-generator predictions of §V-D possible. It must be computed on the noiseless channel: the effective rank of the *estimate* is set by the noise floor and collapses every configuration onto one value.

## IV. PERFORMANCE BENCHMARK

### A. Rank-one Fisher information

Conditional on  $\mathbf{G}, \mathbf{S}, \mathbf{B}$  the noiseless field at  $(n, p)$  is a constant  $\lambda$ , so  $z = |\lambda + w|$  is Rice distributed and its density depends on  $\lambda$  *only through*  $a = |\lambda|$ . With  $\frac{d}{dx} \log I_0(x) = R(x)$  and  $\kappa = 2za/\sigma^2$ , the score is  $\partial_a \log p = \frac{2}{\sigma^2} [zR(\kappa) - a]$ ; writing  $\beta = \sigma^{-4} (\mathbb{E}[z^2 R^2(\kappa)] - a^2)$ , one measurement carries scalar information  $4\beta$ , so in real coordinates  $\boldsymbol{\eta} = [\Re \text{vec } \mathbf{G}; \Im \text{vec } \mathbf{G}]$ ,

$$\mathbf{J}_n = \sum_{p=1}^P 4\beta(a_{n,p}, \sigma^2) \mathbf{u}_{n,p} \mathbf{u}_{n,p}^T. \quad (6)$$

**Each magnitude measurement contributes rank-one information, not rank two.** Crediting both quadratures overstates it and gives a bound too low by 0.17–4.64 dB across our 36 points. An identifiability check settles the form: at  $K = 3$ ,  $P = 3$  the per-row problem is unidentifiable and (6) is correctly singular, whereas the rank-two construction claims full rank.

Because row  $n$  of  $\mathbf{GS}$  enters only measurement row  $n$ ,  $\mathbf{J}$  is *block diagonal across receive elements*. This is the reason an unstructured estimator, and any estimator attaining the unconstrained bound, is flat in  $N$  under a normalised metric — the result §V-B measures.

### B. Constrained bound on the geometric manifold

The unconstrained bound governs estimators treating  $\mathbf{G}$  as  $2NK$  free parameters. HS-GS exploits (2), whose parameters  $\phi \in \mathbb{R}^{3 \sum_k L_k}$  number about 45, *independent of*  $N$ . The text-book form  $\mathbf{D}(\mathbf{D}^T \mathbf{J} \mathbf{D})^{-1} \mathbf{D}^T$  with  $\mathbf{D} = \partial \boldsymbol{\eta} / \partial \phi$  is unusable,  $\mathbf{D}$  being rank deficient: at  $N = 8$  its mean numerical rank is 40.4 against  $\approx 44.7$  parameters, and 28.1% of realisations have  $3 \sum_k L_k > 2NK$ , so  $\phi \mapsto \mathbf{G}$  is not injective. We use the Gorman–Hero / Stoica–Ng tangent-space form [9], [10] with  $\mathbf{U}$  an orthonormal basis of  $\text{range}(\mathbf{D})$ :

$$\text{CCRB} = \mathbf{U}(\mathbf{U}^T \mathbf{J} \mathbf{U})^{-1} \mathbf{U}^T, \quad (7)$$

which depends on the tangent *space* and is invariant to over-parameterisation. Both bounds average over the same realisations as the estimator curves.

**Bias caveat.** Both bounds constrain the covariance of *unbiased* estimators. All estimators here run to a fixed budget with a shrinking update, and HS-GS additionally truncates rank, so none is unbiased and neither bound applies strictly: the unconstrained CRLB is a reference for GS and EM-GS only, and the CCRB does not forbid a sufficiently biased structured estimator from falling below it.

TABLE I  
CONFIGURATIONS. EVERY FIGURE IS INTERNALLY  
SINGLE-CONFIGURATION; NONE MIXES POINTS FROM THE TWO.

|                   | Bound comparisons                       | Effective-rank            |
|-------------------|-----------------------------------------|---------------------------|
| $N$               | 8, 16, 32                               | 16, 32, 64                |
| $K$               | 3                                       | 3                         |
| $P$               | 30 (Figs. 1, 3); 10 and 30 (Fig. 2)     | 20                        |
| $L_k$             | $\mathcal{U}\{3, 7\}$ [5]               | $\mathcal{U}\{3, 7\}$ [5] |
| SNR               | $\{-5, 0, 5, 10, 15, 20\}$ dB           | $\mathcal{U}[-10, 20]$    |
| RSR (single-user) | 12 dB                                   | 10 dB                     |
| Iterations $T$    | 50                                      | 100                       |
| Cadzow sweeps     | 4                                       | 1                         |
| Statistic         | ratio-of-sums                           | paired median, CI         |
| Trials            | 400 per point, $1.44 \times 10^4$ total | 83–874 per cell           |

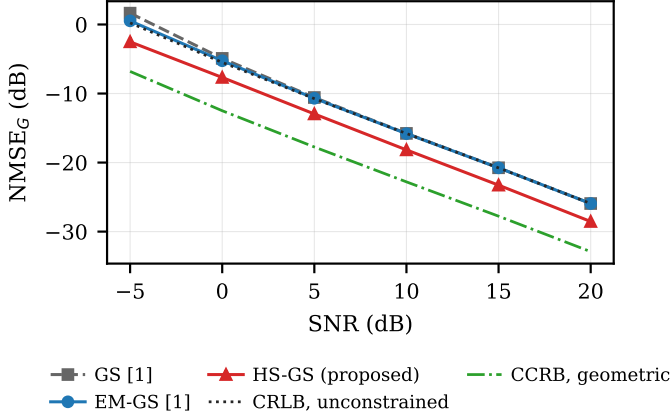


Fig. 1. NMSE versus SNR, bound-comparison configuration ( $N = 32$ ,  $P = 30$ , RSR 12 dB,  $T = 50$ , 4 Cadzow sweeps, ratio-of-sums). EM-GS tracks the rank-one unconstrained CRLB to within 0.051 dB above 5 dB. Neither bound governs a biased estimator.

## V. NUMERICAL RESULTS

Table I lists both configurations. The older is canonical for everything compared to a bound, because it is what the bounds were computed against; the effective-rank and cross-generator results are reported at their own configuration, and each figure names the one it uses.

### A. Main result against the bounds

At  $N = 32$ ,  $P = 30$ , RSR 12 dB (Fig. 1) EM-GS numerically tracks the unconstrained CRLB to within 0.051 dB for  $\text{SNR} \geq 5$  dB. We do not call it efficient: it is biased, and the bound formally governs unbiased estimators (§IV); the close numerical agreement is what makes the benchmark informative. At  $-5$  and  $0$  dB the curves separate by 0.295 and 0.237 dB, consistent with the low-SNR bias of a shrinking update. At this configuration the geometric CCRB lies 7.05–7.11 dB below the unconstrained bound, quantifying what the structure is worth in principle, and HS-GS occupies part of that gap, improving on EM-GS by 2.27–3.04 dB across the sweep.

### B. Aperture scaling

Fig. 2 isolates the scaling. Averaged over the twelve operating points at each array size — the six SNR values of

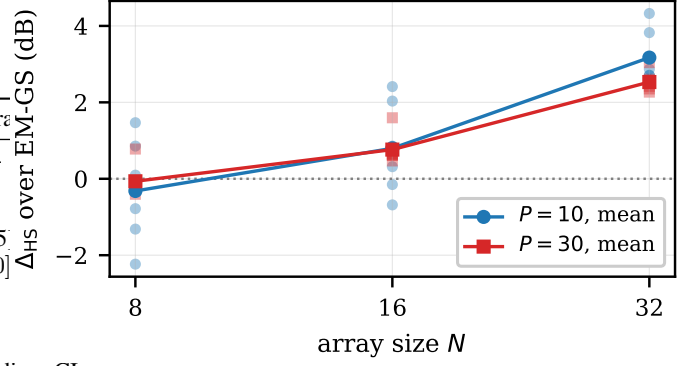


Fig. 2. Paired HS-GS gain over EM-GS versus array size, bound-comparison configuration. Two pilot counts are shown:  $P = 30$ , which Figs. 1 and 3 also use, and  $P = 10$  as an auxiliary curve. Faint markers are the six individual SNR operating points of each curve; solid lines are their means. The headline  $-0.19/+0.78/+2.85$  dB is the mean over all twelve points (six SNR values  $\times$  two pilot counts). At  $N = 8$  the points straddle zero, so no single scalar summarises that array size.

Table I at each of  $P \in \{10, 30\}$  — the gain is  $-0.19, +0.78, +2.85$  dB at  $N = 8, 16, 32$ , with win rates 33.5, 74.0, 95.3% and the constraint active in 57.5, 92.7, 99.6% of trials. **Under the same averaging EM-GS varies by 0.012 dB**, exactly as the block-diagonal Fisher structure of §IV requires: the  $N$ -dependence belongs to the structural prior, not to a degrading baseline.

$N = 8$  is a mixed result, not a small positive one. There  $r_{\max} = 4$  while  $L_k \in \{3, \dots, 7\}$ , so the projection is inactive in 42.5% of trials and where active often truncates real components; HS-GS gains  $+0.78$  and  $+1.47$  dB at  $-5$  dB (at  $P = 30$  and  $P = 10$ ) but turns negative as SNR rises, reaching  $-2.23$  dB, so the sign of any scalar summary depends on the pooling. Truncation bias does not shrink with noise, so as variance falls the trade stops paying.

### C. From path count to effective rank

Fixing  $L_k = L$  across users and sweeping to the ceiling, the gain decays strictly monotonically, 7.04, 3.56, 1.79, 1.04, 0.58, 0.27, 0.05,  $-0.12$  dB for  $L = 2, 4, \dots, 16$ , with EM-GS flat to 0.19 dB across the sweep — which rules out generic denoising, since that would persist at large  $L$  where the estimate is equally noisy rather than vanish exactly where the prior becomes vacuous.

Re-indexing that sweep by  $r_{\text{eff}}/r_{\max}$  generalises it. The zero crossing moves from  $L/r_{\max} = 0.90$  to  $r_{\text{eff}}/r_{\max} = 0.518$ , and the useful region becomes  $r_{\text{eff}}/r_{\max} \lesssim 0.5$  — a statement about any channel rather than about one simulator's  $L$  values. Fig. 3 places  $N \in \{16, 32, 64\}$  on that axis. The crossing is 0.588, 0.518, 0.544 across a fourfold change in aperture, a spread of 0.070; the  $N = 16$  and  $N = 64$  values are extrapolated from the last two measured points rather than bracketed.

**The gain magnitude does not collapse onto the same axis, and we state the residual rather than report a collapse.** Over the window both arrays cover,  $N = 16$  lies above  $N = 64$  by a mean of 0.493 dB and a maximum of 0.901 dB, one-signed throughout and monotone in  $N$ . We have no account

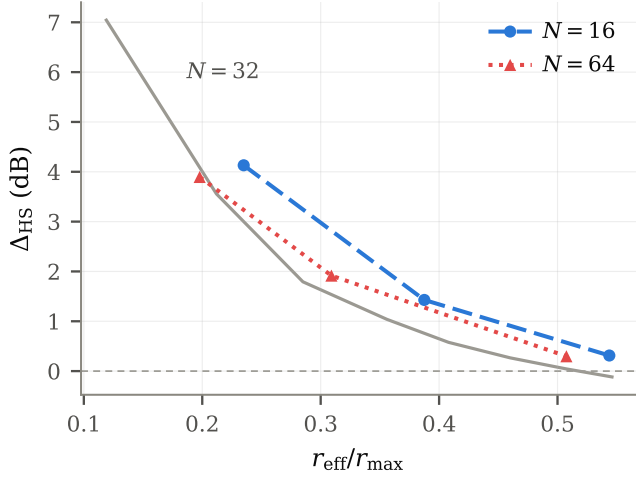


Fig. 3.  $\Delta_{\text{HS}}$  against  $r_{\text{eff}}/r_{\text{max}}$ , effective-rank configuration (Table I, right column; adaptive order, 1 Cadzow sweep, paired median). The boundary is invariant to within 0.070 across  $N \in \{16, 32, 64\}$ ; the magnitude is not, with a one-signed residual of mean 0.493 dB (max. 0.901 dB).

of it. One candidate is pencil quantisation at small capacity —  $r_{\text{max}}(16) = 8$  admits eight ranks against 32 at  $N = 64$  — but that is untested.

#### D. Cross-generator prediction: the falsification test

An earlier stage of this work concluded that robustness under clustered propagation was not established and named the decisive next experiment: *repeat the principal comparison under a clustered generator — a falsification test, not a confirmation exercise*. This is that test, run twice. In each case  $r_{\text{eff}}/r_{\text{max}}$  was computed, a predicted  $\Delta_{\text{HS}}$  read off the relation, and **the prediction committed to version control before the measurement was run**; the relation was not refitted afterwards.

**The prediction rule, stated so it can be reproduced.** The relation is the eight-point table of  $(r_{\text{eff}}/r_{\text{max}}, \Delta_{\text{HS}})$  pairs obtained by re-indexing the path-count sweep of §V-C, and the predicted  $\Delta_{\text{HS}}$  is its *piecewise-linear interpolation* at the measured  $r_{\text{eff}}/r_{\text{max}}$  — no curve is fitted, no regression or spline is involved. The prediction interval is that same interpolation evaluated at the interquartile endpoints of  $r_{\text{eff}}$  over the channel realisations, so it reflects channel variability rather than being asserted. Both the table and the rule were fixed before either measurement below.

**Clustered Saleh–Valenzuela.** At  $r_{\text{eff}}/r_{\text{max}} = 0.331$  the relation predicts +1.30 dB. Measured: +1.173 dB, CI [+1.017, +1.380] — an error of 0.13 dB, prediction inside the interval.

**A second, independently specified configuration.** We adopt the receive-side Saleh–Valenzuela configuration of [11] —  $L = 10$  paths,  $\alpha_\ell \sim \mathcal{CN}(0, 1)$ , angles  $\mathcal{U}(-90^\circ, 90^\circ)$ . That work is a *precoding* study, not a competing estimator, and only its channel configuration is borrowed; it uses ULAs at both ends with incident and departure angles, of which we use the receive side only. At  $r_{\text{eff}}/r_{\text{max}} = 0.400$  the relation predicts

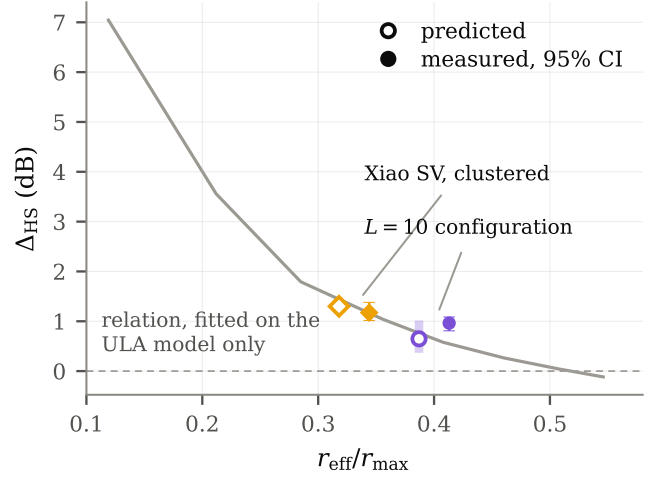


Fig. 4. Predictions registered before measurement, against measurement, effective-rank configuration. Both generators produce channels of the form (2); what differs is the distribution over  $(L_k, \alpha_{k\ell}, \theta_{k\ell})$ . The relation was fitted on the generator of §V alone and not refitted afterwards.

+0.648 dB, interval [+0.367, +1.015]. Measured: +0.963 dB, CI [+0.808, +1.086]. Inside the registered interval and on the correct side of zero, but the point estimate is 0.315 dB low — a larger error than the first, and we report it as such rather than as a second 0.13 dB hit. Together the two bound what the relation is good for: it places an unseen channel on the correct part of the curve, and it is not accurate to a tenth of a decibel in general.

**Clustering did not eliminate the gain**, which was the outcome the earlier stage identified as possible. On a third reading in which all 40 rays are drawn independently,  $r_{\text{eff}}/r_{\text{max}}$  rises to 0.748, the relation predicts −0.12 dB, and the measured median is exactly 0.000 dB. The magnitude is right to 0.12 dB but *the mechanism is not the predicted one*: rather than degrading, the order-selection rule declines to project at all in 29.1% of trials against 1.4% on the clustered model, without being told which channel it faces. We report that as a partial match.

#### E. Users, and a replication

The  $K$ -invariance predicted by (5) holds at  $\text{SNR} \geq 5$  dB, where  $\Delta_{\text{HS}}$  across  $K \in \{2, 3, 4\}$  at fixed  $P/2K$  spans 0.095 dB. Pooled over all SNR it does not: the spread is 0.294 dB and monotone in  $K$ , missing a pre-registered falsifier of 0.3 dB by 0.006 dB, which we do not present as a pass. The pooled trend is driven by bins below 0 dB where each cell contributes about 45 trials.

**The two configurations are a replication.** At  $N = 32$  the same contrast measures +2.85 dB under the bound-comparison configuration and +2.680 dB under the effective-rank one — different code, different RSR, iteration count, Cadzow sweep count and pooling rule, agreeing to 0.17 dB.

## VI. LIMITATIONS AND CONCLUSION

A sparse geometric ULA channel makes each column's Hankel lifting rank deficient, and this can be enforced inside

an exact biased phase-retrieval iteration without linearising the magnitude observation or estimating any angle. Because the Fisher information is block diagonal across receive elements under the unconstrained row-wise parameterisation, the unstructured baselines are flat in array size while HS-GS converts a larger aperture into lower normalised error:  $-0.19 \rightarrow +2.85$  dB as  $N$  goes  $8 \rightarrow 32$  against 0.012 dB of baseline movement. The governing variable is effective rank relative to capacity, not path count, and that relation predicted the gain on two unseen channel-generation distributions — one to within 0.13 dB, the other inside a registered interval but 0.32 dB off. Clustered propagation, identified earlier as the decisive falsification test, did not eliminate the gain.

**Limitations.** All results are simulation only, with no measured hardware and no recorded channel data. They cover the geometric ULA model of (2) and the Saleh–Valenzuela configurations of §V-D, and no claim is made beyond those families; the gain magnitude does not collapse onto the proposed axis and its residual is unexplained; the benefit is conditional and negative at  $N = 8$  above 0 dB. A single scalar order is imposed on all  $K$  users despite their differing path counts, the projection step is approximate on a non-convex set with no convergence guarantee, and equal user power leaves near-far effects untested. Finally, both bounds constrain unbiased estimators, and every estimator here is biased.

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